

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250275887

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Jeppsson; Anders et al.

MECHANICAL CARDIO PULMONARY RESUSCITATION DEVICE HAVING A CONTACT MEMBER

Abstract

A mechanical CPR device having one or more of a piston, a driving component configured to extend the piston toward a patient's torso and retract the piston away from the patient's torso, a controller configured to control the driving component to at least compress the patient's torso by extending the piston from a reference position to a depth and retracting the piston from the depth to the reference position, and a contact member such as one or more of a pressure pad and a suction cup attached to the end of the piston. The contact member can include a semi-adhesive material that has low adhesiveness when the controller controls the driving component to compress the patient's torso less than 60 times per minute and high adhesiveness when the controller controls the driving component to compress the patient's torso more than 60 times per minute.

Inventors: Jeppsson; Anders (Lund, SE), Chapman; Fred W. (Newcastle, WA)

Applicant: PHYSIO-CONTROL, INC. (Redmond, WA)

Family ID: 80356122

Assignee: PHYSIO-CONTROL, INC. (Redmond, WA)

Appl. No.: 19/210501

Filed: May 16, 2025

Related U.S. Application Data

parent US continuation 17460614 20210830 parent-grant-document US 12329714 child US 19210501

us-provisional-application US 63074033 20200903

Publication Classification

Int. Cl.: A61H31/00 (20060101); **A61H9/00** (20060101)

U.S. Cl.:

CPC **A61H31/006** (20130101); **A61H9/005** (20130101); A61H2009/0064 (20130101);
A61H2201/105 (20130101); A61H2201/1664 (20130101); A61H2201/5007 (20130101)

Background/Summary

PRIORITY [0001] This application is a continuation of U.S. Non-Provisional patent application Ser. No. 17/460,614, filed Aug. 30, 2021, which claims priority to U.S. Provisional Application No. 63/074,033, filed Sep. 3, 2020, the disclosures of both of which are incorporated herein by reference in their entirety.

FIELD

[0002] The present disclosure relates to a system and method of Cardio Pulmonary Resuscitation (CPR) including a CPR chest compression machine having a contact member.

BACKGROUND

[0003] In certain types of medical emergencies a patient's heart stops working. This stops the blood flow, without which the patient may die. Cardio Pulmonary Resuscitation (CPR) can forestall the risk of death. CPR includes performing repeated chest compressions to the chest of the patient so as to cause their blood to circulate some. CPR also includes delivering rescue breaths to the patient. CPR is intended to merely maintain the patient until a more definite therapy is made available, such as defibrillation. Defibrillation is an electrical shock deliberately delivered to a person in the hope of correcting their heart rhythm.

[0004] Guidelines by medical experts such as the American Heart Association provide parameters for CPR to cause the blood to circulate effectively. The parameters are for aspects such as the frequency of the compressions, the depth that they should reach, and the full release that is to follow each of them. The depth is sometimes required to exceed 5 centimeters (cm) (2 inches (in.)). The parameters also include instructions for the rescue breaths.

[0005] Traditionally, CPR has been performed manually. A number of people have been trained in CPR, including some who are not in the medical professions just in case. However, manual CPR might be ineffective, and being ineffective it may lead to irreversible damage to the patient's vital organs, such as the brain and the heart. The rescuer at the moment might not be able to recall their training, especially under the stress of the moment. And even the best trained rescuer can become quickly fatigued from performing chest compressions, at which point their performance might be degraded. Indeed, chest compressions that are not frequent enough, not deep enough, or not followed by a full decompression may fail to maintain blood circulation.

[0006] The risk of ineffective chest compressions has been addressed with CPR chest compression machines. Such machines have been known by a number of names, for example CPR chest compression machines (CCCM), mechanical CPR devices, cardiac compressors and so on.

[0007] CPR chest compression machines repeatedly compress and release the chest of the patient. Such machines can be programmed so that they will automatically compress and release at the recommended rate or frequency, and can reach a specific depth within the recommended range. The repeated chest compressions of CPR are actually compressions alternating with releases of a compression element, such as a piston (also referred to as a plunger) or belt. Conventional CPR machines start from a starting point (also referred to as a reference point), apply the chest compression, then release the compression element back to its starting point to reset and be ready to apply another chest compression, when needed, or according to a protocol. The start position of

the compression element is near or physically touching the patient's chest.

[0008] Some CPR machines can even exert force upwards during the “release” or decompressions, pulling the chest higher than it would be while at rest—a feature that is called active decompression. Active decompression applies a force to promote the patient's chest to expand after an applied chest compression. During active decompression in examples of a CCCM with a piston with a suction cup, the suction cup creates a vacuum with the patient's chest and applies a force directed away from the anterior surface of the patient's chest to promote chest expansion beyond the chest's starting resting height between chest compressions.

[0009] The correct positioning of a mechanical CPR device on a patient's chest is critical due to intended use. In examples of a CCCM with a piston with a contact member such as a pressure pad, a suction cup, or a belt, the contact member distributes the chest compression force from the CPR device to the patient. The contact member may slide or disconnect during mechanical CPR. The reasons for sliding or disconnecting may be chest skin hairiness or the outer shape of the chest. There is a risk that sliding or disconnecting is gradually taking place without the awareness of the user. A sliding contact member may cause injury to the patient. Although big shifts in placement are easily seen, it can be difficult to see that the contact member has shifted small amounts. Importantly, even a small shift toward the head can move the point of compression to be over the left-ventricular outflow tract, where it can greatly impede the desired forward flow of blood. Such a shift needs to be detected and corrected promptly. A partly or fully disconnected contact member may take away the ability to provide active decompression or other intended treatments.

[0010] A firm fixation of the contact member of a CCCM to the patient's chest is essential during active decompression. If the contact member is not firmly attached to the patient's chest undesired effects may occur. For example, if the CCCM continues to lift the contact member, without lifting the patient's chest, the chest may get sore and left with abrasions as the contact member will hit the patient's chest after each lift of the contact member. Furthermore, if the CCCM stops the active decompression the patient will not receive active decompression when desired.

[0011] In examples of a CCCM with a piston with a suction cup, the vacuum created between the patient's chest and the suction cup makes it difficult for the suction cup to move laterally from the desired compression area and allows for active decompression as the suction cup can transfer traction. However, the functionality of the suction cup with regards to transferring traction is dependent on a good connection to the patient's chest as it is the low pressure created in the suction cup that contributes to the lifting force. If the connection to the patient's chest is not favorable due to i.e. hairy skin, other properties of the patient's skin or the topography of the chest in general, the suction cup and the CCCM may lose its ability to perform active decompression.

SUMMARY

[0012] In some examples, the present disclosure includes a mechanical CPR device having one or more of a piston, a driving component configured to extend the piston toward a patient's torso and retract the piston away from the patient's torso, a controller configured to perform mechanical CPR by controlling the driving component to at least compress the patient's torso by extending the piston from a reference position to a depth and retracting the piston from the depth to a reference position, wherein the reference position is the position from which the depth of CPR compressions are measured, and a pressure pad attached to the end of the piston. The pressure pad can have a pressure pad contact surface area that includes a material disposed on the pressure pad contact surface. The material can be configured to attach to a patient target area and can include a semi-adhesive material that has low adhesiveness when the controller controls the driving component to compress the patient's torso less than 60 times per minute and high adhesiveness when the controller controls the driving component to compress the patient's torso more than 60 times per minute. Additionally or alternatively, the material includes ink configured to mark an initial contact location on the patient's torso.

[0013] Additionally or alternatively, the CPR device can include a pressure pad protective layer

disposed on the pressure pad such that the material is disposed between the pressure pad contact surface area and the pressure pad protective layer. Additionally or alternatively, the controller is further configured to actively decompress the patient's torso by retracting the piston from the reference position to a height above the reference position, whereby the patient's torso is decompressed above the torso's natural resting position and above the reference position. Additionally or alternatively, the CPR device can also include a suction cup attached to the end of the piston, the suction cup having a suction cup contact surface configured to attach to the patient's torso, the pressure pad disposed within the suction cup and not in contact with the suction cup contact surface.

[0014] Alternative examples of the present disclosure can include a mechanical CPR device including a piston, a driving component configured to extend the piston toward a patient's torso and retract the piston away from the patient's torso, a controller configured to perform mechanical CPR by controlling the driving component to at least compress the patient's torso by extending the piston from a reference position to a depth and retracting the piston from the depth to a reference position, wherein the reference position is the position from which the depth of CPR compressions are measured, and a suction cup attached to the end of the piston. The suction cup can have a suction cup contact surface area and a material disposed on the contact surface area, the material configured to attach to a target area on a patient chest. Additionally or alternatively, the material includes ink configured to mark an initial contact location on the patient's torso.

[0015] Additionally or alternatively, the CPR device can include a suction cup protective layer disposed on the suction cup such that the material is disposed between the suction cup contact surface area and the suction cup protective layer. Additionally or alternatively, the material includes a semi-adhesive material that has low adhesiveness when the controller controls the driving component to compress the patient's torso less than 60 times per minute and high adhesiveness when the controller controls the driving component to compress the patient's torso more than **60** times per minute. Additionally or alternatively, the CPR device can include a pressure pad attached to the end of the piston, the pressure pad disposed within the suction cup and not in contact with the suction cup contact surface.

[0016] A method of attaching a suction cup to a patient's torso, the suction cup located on an end of a piston of a mechanical CPR device and having a suction cup contact area, in accordance with some examples of the present disclosure can include extending, by the mechanical CPR device, the piston until a first position at which the suction cup comes into contact with the patient's torso. Further extending, by the mechanical CPR device, the piston to cause air to be forced out from an area between the suction cup and the patient's torso, and removing a protective layer disposed between the suction cup contact area and the patient's torso.

[0017] These and other features and improvements of the present application and the resultant patent will become apparent to one of ordinary skill in the art upon review of the following detailed description when taken in conjunction with the several drawings and the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] FIG. 1 is a schematic block diagram of an exemplary schematic block diagram of a mechanical CPR device in accordance with the present disclosure.

[0019] FIG. 2 is an exemplary CPR system including a piston and a contact member in accordance with the present disclosure.

[0020] FIG. 3 is a partial view of a CPR device showing a piston and pressure pad.

[0021] FIG. 4 is an underside view of the piston and pressure pad of FIG. 3 further including an exemplary protective layer.

[0022] FIG. 5 is a perspective view of a suction cup in accordance with the present disclosure.

[0023] FIG. 6 is a perspective view of the suction cup of FIG. 5 further including a protective layer in accordance with the present disclosure.

[0024] FIG. 7 is an exemplary flow chart of attachment of a CPR device including a piston and a suction cup in accordance with the present disclosure.

DETAILED DESCRIPTION

[0025] As disclosed herein, examples are directed to a mechanical CPR device having a contact member including a material disposed on a contact member surface. The material can include an adhesion material to increase the sealing effect between a target contact area of the patient's skin and the contact member such that the functionality of the contact member remains and the adherence improved. Additionally or alternatively, the material can include ink such that the patient's torso is automatically marked during a CPR cycle to make it apparent if the contact member has migrated away from a target contact area on the patient's torso.

[0026] FIG. 1 illustrates an example schematic block diagram of a mechanical CPR device **100**. As will be understood by one skilled in the art, the mechanical CPR device **100** may include additional components not shown in FIG. 1. The mechanical CPR device **100** includes a controller **102**, which may be in electrical communication with a chest compression mechanism or device **104**. The chest compression mechanism **104** can include a compression element that compresses a chest of a patient, such as a piston based chest compression device or a belt driven device that wraps around a chest of a patient, and an optional contact member, such as a pressure pad, a suction cup, and/or a belt. The compression element in FIG. 1 includes a piston **106** and a contact member **154**. Chest compression elements for CPR machines can also include compression arms, such as one or more rigid or semi-rigid arms and/or a compression element and belt combination. The rigid or semi-rigid arms apply a force onto the anterior surface of the patient's chest in a manner similar to that of the piston-style chest compression element. The belt-style chest compression element is often a flexible but resilient and tough material that tightens around some portion of the patient's chest to force its compression element to against the patient's chest to apply the chest compression. Alternatively, the belt holds a compression element, such as a plunger or piston element above the patient's chest and tightens the belt on either side of the plunger or position to cause the force to move the plunger or piston toward the patient's chest. The belt can be made of any suitable material.

[0027] Contact member **154** can include a suction cup, a compression pad, a suction cup including a compression pad, a belt, or other device configured to make contact with a patient's chest. The chest compression mechanism **104** can further include a contact surface **116** configured to make contact with a patient's chest. The contact surface **116** can be disposed on the piston **106** or the contact member **154**. The chest compression mechanism **104** further can include retention structure **108** including one or more legs **110** and/or a support portion **112** configured to be placed underneath a patient **114**.

[0028] The chest compression mechanism **104** may include a driver **118** configured to drive the compression mechanism **104** to cause the compression mechanism **104** to perform compressions to a chest of patient **114**. The controller **102** provides instructions to the chest compression mechanism **104** to operate the chest compression mechanism **104** at a number of different rates, depths, heights, duty cycles.

[0029] The controller **102** may include a processor **120**, which may be implemented as any processing circuitry, such as, but not limited to, a microprocessor, an application specific integration circuit (ASIC), programmable logic circuits, etc. The controller may further include a memory **122** coupled with the processor **120**. Memory can include a non-transitory storage medium that includes programs **124** configured to be read by the processor **120** and be executed upon reading. The processor **120** is configured to execute instructions from memory **122** and may perform any methods and/or associated operations indicated by such instructions. Memory **122** may be

implemented as processor cache, random access memory (RAM), read only memory (ROM), solid state memory, hard disk drive(s), and/or any other memory type. Memory **122** acts as a medium for storing data **126**, such as event data, patient data, etc., computer program products, and other instructions.

[0030] Controller **102** may further include a communication module **128**. Communication module **128** may transmit data to a post-processing module **130**. Alternately, data may also be transferred via removable storage such as a flash drive. While in module **130**, data can be used in post-event analysis. Such analysis may reveal how the CPR machine was used, whether it was used properly, and to find ways to improve future sessions, etc.

[0031] Communication module **128** may further communicate with other medical device **132**. Other medical device **132** can be a defibrillator, a monitor, a monitor-defibrillator, a ventilator, a capnography device, or any other medical device. Communication between communication module **128** and other medical device **132** could be direct, or relayed through a tablet or a monitor-defibrillator. Therapy from other device **132**, such as ventilation or defibrillation shocks, can be coordinated and/or synchronized with the operation of the CPR machine. For example, compression mechanism **104** may pause the compressions for delivery of a defibrillation shock, afterwards detection of ECG, and the decision of whether its operation needs to be restarted. For instance, if the defibrillation shock has been successful, then operation of the CPR machine might not need to be restarted. Additionally or alternatively, the other medical device **132** can include a ventilator and the ventilator can send instructions to the controller **102** to coordinate chest compressions and ventilation.

[0032] The controller **102** may be located separately from the chest compression mechanism **104** and may communicate with the chest compression mechanism **104** through a wired or wireless connection **134**. The controller **102** also electrically communicates with a user interface **136**. As will be understood by one skilled in the art, the controller **102** may also be in electronic communication with a variety of other devices, such as, but not limited to, another communication device, another medical device, etc.

[0033] The chest compression mechanism **104** may include one or more sensors configured to transmit information to controller **102**. For example, chest compression mechanism **104** can include a physiological parameter sensor **138** for sensing a physiological parameter of a patient and to output a physiological parameter sensor signal **140** that is indicative of a dynamic value of the parameter. The physiological parameter can be an Arterial Systolic Blood Pressure (ABSP), a blood oxygen saturation (SpO₂), a ventilation measured as End-Tidal CO₂ (ETCO₂), a temperature, a detected pulse, etc. In addition, this parameter can be what is detected by defibrillator electrodes that may be attached to patient, such as ECG and impedance.

[0034] Additionally or alternatively, the chest compression mechanism **104** can include a height sensor **142** configured to sense the height of the patient's chest and to output a height signal **144**, which is indicative of the resting height of the patient's chest. Additionally or alternatively, the controller **102** can receive the height signal **144** and calculate a reference position, also referred to as a start position, for the compression mechanism **104**. Additionally or alternatively, the chest compression mechanism can include a movement sensor **146** configured to sense movement of the patient's chest and to output a movement signal **148**, which may indicate ventilation movement of the patient's chest. Additionally or alternatively, the chest compression mechanism **104** can include a pressure sensor **150** configured to sense area(s) of pressure of the contact surface with the patient's chest and to output a pressure signal **152**, which is indicative of a dynamic value of pressure against the patient's chest.

[0035] Operations of the mechanical CPR device **100** may be effectuated through the user interface **136**. The user interface **136** may be external to or integrated with a display. For example, in some examples, the user interface **136** may include physical buttons located on the mechanical CPR device **100**, while in other examples, the user interface **136** may be a touch-sensitive feature of a

display. The user interface **136** may be located on the mechanical CPR device **100**, or may be located on a remote device, such as a smartphone, tablet, PDA, and the like, and is also in electronic communication with the controller **102**.

[0036] During a CPR session of compressions, controller **102** can generate or receive an instruction (either pre-programmed or customized based on any parameters or other data) to drive the compression mechanism **104** from a reference position towards the patient's chest to a compression position to administer a chest compression. The reference position can be a specific and pre-defined position or can be calculated or estimated based on sensed input or other patient and/or rescuer data. The same or a subsequent instruction can also drive the compression mechanism **104** to move back away from the patient's chest after the applied chest compression.

[0037] FIG. **2** shows a CPR system **200** including a retention structure **202**. The retention structure **202** includes a central member **204**, a first leg **206**, a second leg **208**, and a support portion **210** configured to be placed underneath a patient. Central member **204** is coupled with first leg **206** and with second leg **208** via joints **214** and **214**, respectively. In addition, the far ends of legs **206**, **208** can become coupled with edges **216**, **218** of support portion **210**. These couplings form the retention structure **202** that retains a patient. In this particular case, central member **204**, first leg **206**, second leg **208** and support portion **210** form a closed loop, in which the patient is retained.

[0038] Central member **204** includes a battery that stores energy, a motor that receives the energy from the battery, and a compression mechanism that can be driven by the motor. The compression mechanism is driven up and down by the motor using a rack and pinion gear. The compression mechanism includes a compression element, such as a piston **220** that emerges from central member **204**, and can compress and release the patient's chest. Piston **220** is sometimes called a plunger. Here, piston **220** terminates in a contact member **222** having a contact surface **224**. The contact member **222** can include a pressure pad, a suction cup, or a suction cup including a pressure pad, a belt, or other device configured to contact a patient chest. In the example shown in FIG. **2**, the contact member **222** includes a suction cup. Here, the battery, the motor and the rack and pinion gear are not shown, because they are completely within a housing of central member **204**.

[0039] FIG. **3** is a partial view of a CPR device **300** including a piston **302** and a contact member **304**. The contact member **304** in FIG. **3** includes a pressure pad **306**. The pressure pad **306** has a pad contact surface **308** including a material **310**. The material **310** may cover all or a portion of the pad contact surface **308**.

[0040] In some examples, the material **310** is configured such that the piston **302** of the CPR device **300** stays in place with respect to a targeted contact area on a patient's chest skin during treatment. In such examples, the material **310** causes the pressure pad **306** to stick to the targeted contact area and prevents the pressure pad **306** from sliding on the patient's chest. Additionally or alternatively, the material **310** can connect the patient's skin to the CPR device **300** in such way that active decompression can be provided regardless of target contact area impurities such as hair or the shape of the patient's chest.

[0041] The material **310** may include a semi-adhesive material that is adhesive when in dynamic use and/or has low-to-no adhesiveness when in static use. In other words, the properties of the semi-adhesive material is such that it is strongly adhesive when in dynamic use, such as in use of with a mechanical CPR device where the chest is compressed over 60 times per minute or about 100 times per minute. Additionally or alternatively, the semi-adhesive material adhesiveness is low when in static use, such as when the treatment has been completed and the contact pad is to be removed from the patient or where the chest is compressed less than 60 times per minute.

Accordingly, when pulling pad contact surface **308** including the semi-adhesive material manually from the skin with a low and static force, the semi-adhesive material will loosen and come off the patient's skin. The semi-adhesive material can also be referred to as a non-linear adhesive material.

[0042] Additionally or alternatively, in some examples the material **310** is configured such that a target area of a patient's chest is automatically marked once the patient's chest is contacted by the

material **310** to make it apparent if the pressure pad **306** has migrated away from its initial position. For example, the material **310** can include ink that transfers to patient skin at an initial contact location once the pressure pad **306** is in contact with the patient. In some examples, the ink can transfer to the patient skin during the first compression. In an alternative example, the ink could be visible only under a certain kind of light, for example a black light. An LED of the correct wavelength can be provided on the patient facing side of the CPR device's central member or other patient facing portion of the CPR device. This would reduce the messy appearance of the ink on the chest.

[0043] FIG. **4** shows the underside of the pressure pad **306** of the partial view of the CPR device **300** of FIG. **3**. FIG. **4** further shows a pad protection layer **312** disposed on the pad contact surface **308** including the material **310**. The pad protection layer **312** may cover all or some of the pad contact surface **308**. The pad protection layer **312** can include a pull tab **314**, also referred to as a handle. The material **310** can be activated by a user after the mechanical CPR device **300** with pressure pad **304** has been correctly positioned at a target contact area on a patient's chest by removing the pad protection layer **312** via the pull tab **314**. After removal of the pad protection layer **312**, the chest compression can begin and the material **310** comes in contact with the patient's chest. Once in contact with the patient's skin, the material **310** can adhere the pressure pad **304** to the patient's chest as described above and/or mark the target area of the patient's chest during the first compression.

[0044] A pressure pad having a pad contact surface including a material can be used alone or in conjunction with a suction cup as discussed in further detail below. The pressure pad may be a single use accessory to be exchanged before each treatment of a patient. In some examples the pressure pad can be approximately 5 centimeters in diameter and/or include a height of approximately 1 centimeter. The size of the pressure pad may depend on the adhesive properties of the adhesion material and/or the homogeneity/density of the pressure pad material. For example, a pressure pad having an adhesion material having higher adherence may be smaller in diameter. With respect to thickness, the pressure pad includes some adaption to the chest such that the pad is able to attach to the skin despite irregularities due to the shape of the rib cage. However, a pressure pad that is overly thick would be less stiff and the device CPR device would need to compensate for the compression and elongation of the pad during compressions by increasing the stroke length.

[0045] FIG. **5** is a perspective view of a suction cup **400** for use with a CPR machine in accordance with the present disclosure. The suction cup **400** has a suction cup contact surface **402** along the perimeter of the underside of the suction cup. The suction cup contact surface can include a material **404** covering all or a portion of the suction cup contact surface. In some examples, the suction cup **400** can be single use.

[0046] In some examples, the material **404** is configured to maintain connection of the suction cup contact surface **402** to a target contact area of the patient's chest and/or maintain the CPR device in place on a compression area on the patient's chest using for example, an adhesive. For example, in some examples, the material **404** can include a semi-adhesive material or non-linear adhesive material as discussed above with reference to FIGS. **3** and **4**. The material **404** increases the sealing effect between the target contact area of the patient's skin and the suction cup contact surface **402** such that the functionality of the suction cup remains and the adherence improved. In use, the air inside the suction cup **400** is pressed out to reduce the internal volume and create a lower internal air pressure. The material **404** disposed on the suction cup contact area **402** ensures that the adherence between the suction cup contact surface **402** and the patient's skin is high to eliminate air from returning into the suction cup **400**. The material **404** is configured to provide a strong connection between the suction cup **400** and the target contact area of the patient's chest and/or is configured to provide for easy detachment of the suction cup **400** from the patient's chest after completion of the compressions or when there is a need for adjustment or possibly re-adjustment of the position of the suction cup **400**.

[0047] Additionally or alternatively, in some examples the material **404** is configured such that a target area of a patient's chest is automatically marked once the patient's chest is contacted by the material **404** to make it apparent if the suction cup **400** has migrated away from its initial position. For example, the material **404** can include ink that transfers to patient skin at an initial contact location once the suction cup **400** is in contact with the patient. In some examples, the ink can transfer to the patient skin during the first compression. In an alternative example, the ink could be visible only under a certain kind of light, for example a black light. An LED of the correct wavelength can be provided on the patient facing side of the CPR device's central member or other patient facing portion of the CPR device. This would reduce the messy appearance of the ink on the chest.

[0048] FIG. **6** shows a different perspective view of the suction cup **400** of FIG. **5** having a suction cup contact surface **402**. In the example shown in FIG. **6**, the CPR device further includes a suction cup protective layer **406** having a tab **408** disposed over the suction cup contact surface **404**. As shown in FIG. **7**, in use, the protective layer initially separates the material from the patient's skin such that air can be pressed out of the suction cup by lowering the suction cup against the patient (**702**). Once the internal volume of the suction cup is decreased, the protective layer can be removed by a user by pulling a tab (**704**). When the protective layer is removed, the material comes in contact with the patient's skin (**706**) during a first compression. The material then adheres to the patient's chest and/or transfers ink to the patient's chest at the contact location.

[0049] In use, the air inside the suction cup is pressed out to reduce the internal volume and create a lower internal air pressure. The material disposed on the suction cup contact area can ensure that the adherence between the suction cup contact surface and the patient's skin is high to eliminate air from returning into the suction cup and/or can mark the patient's torso at the point of contact. The material can be configured to provide a strong adhesive connection between the suction cup and the target contact area of the patient's chest and/or is configured to provide for easy detachment of the suction cup from the patient's chest after completion of the compressions or when there is a need for adjustment or possibly re-adjustment of the position of the suction cup. In some examples, the material can include a semi-adhesive material or non-linear adhesive material as discussed above with reference to FIGS. **3-6**.

[0050] In some examples, the material can include ink and can be disposed within the suction cup. The material could automatically be released in the first few compressions by having it be pumped out by the positive-negative pressure cycling of those compressions and releases.

[0051] In some examples of a mechanical CPR device, a contact member includes a suction cup attached to the end of a piston. The suction cup has a suction cup contact surface configured to attach to the patient's torso and a material disposed on the suction cup contact surface as previously described. The contact member can further include a pressure pad. The pressure pad can be disposed within the suction cup such that it is not in contact with the suction cup contact surface. The pressure pad includes a pressure pad contact area configured to contact the patient's torso and can include a material disposed on the pressure pad contact area as previously described.

[0052] Although CPR devices including a pressure pad and/or a suction cup are pictured, the disclosure includes examples of a CPR device having a band configured to squeeze the chest. A material can be disposed on a patient facing surface of the band. The material can include an adhesive or semi-adhesive as described above. Additionally or alternatively, the material can include ink to mark an initial contact area of the chest.

[0053] Examples may operate on a particularly created hardware, on firmware, digital signal processors, or on a specially programmed general purpose computer including a processor operating according to programmed instructions. The terms "controller" or "processor" as used herein are intended to include microprocessors, microcomputers, ASICs, and dedicated hardware controllers. One or more aspects may be embodied in computer-usable data and computer-executable instructions, such as in one or more program modules, executed by one or more

computers (including monitoring modules), or other devices. Generally, program modules include routines, programs, objects, components, data structures, etc. that perform particular tasks or implement particular abstract data types when executed by a processor in a computer or other device. The computer executable instructions may be stored on a non-transitory computer readable medium such as a hard disk, optical disk, removable storage media, solid state memory, RAM, etc. As will be appreciated by one of skill in the art, the functionality of the program modules may be combined or distributed as desired in various examples. In addition, the functionality may be embodied in whole or in part in firmware or hardware equivalents such as integrated circuits, field programmable gate arrays (FPGA), and the like. Particular data structures may be used to more effectively implement one or more aspects of the disclosed systems and methods, and such data structures are contemplated within the scope of computer executable instructions and computer-usable data described herein.

[0054] The previously described versions of the disclosed subject matter have many advantages that were either described or would be apparent to a person of ordinary skill. Even so, all of these advantages or features are not required in all versions of the disclosed apparatus, systems, or methods.

[0055] Additionally, this written description makes reference to particular features. It is to be understood that the disclosure in this specification includes all possible combinations of those particular features. For example, where a particular feature is disclosed in the context of a particular aspect or example, that feature can also be used, to the extent possible, in the context of other aspects and examples.

[0056] Also, when reference is made in this application to a method having two or more defined steps or operations, the defined steps or operations can be carried out in any order or simultaneously, unless the context excludes those possibilities.

[0057] Furthermore, the term “comprises” and its grammatical equivalents are used in this application to mean that other components, features, steps, processes, operations, etc. are optionally present. For example, an article “comprising” or “which comprises” components A, B, and C can contain only components A, B, and C, or it can contain components A, B, and C along with one or more other components.

[0058] Also, directions such as “vertical,” “horizontal,” “right,” and “left” are used for convenience and in reference to the views provided in figures. But the CPR device may have a number of orientations in actual use. Thus, a feature that is vertical, horizontal, to the right, or to the left in the figures may not have that same orientation or direction in actual use.

[0059] Although specific examples have been illustrated and described for purposes of illustration, it will be understood that various modifications may be made without departing from the spirit and scope of the disclosure. Accordingly, the invention should not be limited except as by the appended claims.

Claims

1. A mechanical cardiopulmonary resuscitation (CPR) device, comprising: a piston; a driving component configured to extend the piston toward a patient's torso and retract the piston away from the patient's torso; a controller configured to perform mechanical CPR by controlling the driving component to at least compress the patient's torso by extending the piston from a reference position to a compression depth and retracting the piston from the compression depth to the reference position, wherein the reference position is the position from which the depth of CPR compressions are measured; and a contact member attached to the end of the piston, the contact member having a contact surface area and an ink disposed on a portion of the contact surface area, the ink configured to transfer from the contact surface area to the patient's skin to mark an initial contact location on the patient's torso.

2. The mechanical CPR device of claim 1, in which the ink is configured to transfer from the contact surface area to the patient's skin when the driving component is controlled to perform a first compression.
 3. The mechanical CPR device of claim 1, in which the ink is visible only under a black light.
 4. The mechanical CPR device of claim 1, further comprising a protective layer disposed on the contact member such that the ink is disposed between the contact surface area and the protective layer.
 5. The mechanical CPR device of claim 4, in which the protective layer includes a tab configured to be pulled by a user.
 6. The mechanical CPR device of claim 5, in which pulling the tab removes the protective layer and causes the ink to be transferred from the contact surface area to the patient's skin.
 7. The mechanical CPR device of claim 1, in which the contact member comprises a pressure pad.
 8. The mechanical CPR device of claim 1, in which the contact member comprises a suction cup configured to attach to the patient's torso.
 9. The mechanical CPR device of claim 8, in which the contact member further comprises a pressure pad disposed within the suction cup and not in contact with the suction cup contact surface.
 10. The mechanical CPR device of claim 1, further comprising a semi-adhesive material disposed on the contact surface area, the semi-adhesive material configured to attach the contact member to the marked initial contact location on the patient's torso.
 11. A method of tracking migration of a contact member located on an end of a piston of a mechanical CPR device and having a contact area, the method comprising: extending, by the mechanical CPR device, the piston until a first position at which the contact member comes into contact with the patient's torso; and removing a protective layer disposed between the contact area and the patient's torso, wherein an ink is disposed on the contact area, and wherein removing the protective layer transfers the ink to the patient's torso to mark an initial contact location.
 12. The method of claim 11, further comprising viewing the mark of the initial contact location with a black light of the mechanical CPR device.
 13. The method of claim 11, in which removing the protective layer comprises pulling a tab on the protective layer.
 14. The method of claim 11, further comprising extending and retracting the piston to perform CPR compressions to the patient's torso at a compression location, and further comprising determining whether the compression location differs from the marked initial contact location.
 15. The method of claim 11, in which the contact member is a pressure pad.
 16. The method of claim 11, in which the contact member is a suction cup configured to attach to the patient's torso.
 17. The method of claim 16, further comprising attaching the suction cup to the marked initial contact location on the patient's torso with a semi-adhesive material disposed on the contact area.
 18. The method of claim 16, further comprising actively decompressing the patient's torso by retracting a piston from a reference position to a decompression height after the protective layer is removed.
-